

Groundwater — Flow Theory

Exercise 1: Boundary Conditions and Darcy Flow Analysis

The figure below shows a cross-sectional view of a confined aquifer system with various boundary conditions. This problem involves determining the type of boundary conditions present and calculating the Darcy velocity and volumetric discharge rate from the aquifer to a stream.

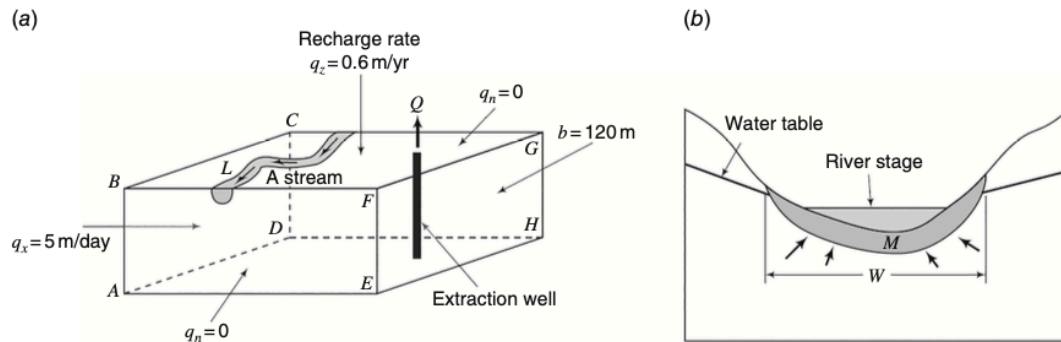


Figure 1: Cross-sectional view of confined aquifer with boundary conditions and extraction well

Given data:

- Length of system: $L = 500$ m
- Width of system: $W = 20$ m
- Aquifer thickness: $M = 0.5$ m
- Hydraulic conductivity: $K_m = 0.005$ m/day
- Head at stream: $h = 20$ m
- Head at monitoring point: $h_m = 18$ m

System characteristics from the figure:

- Constant head boundary on the right side: $h = 120$ m
- No-flow boundaries on front and back sides: $q_n = 0$
- Inflow velocity on the left side: $q_x = 5$ m/day
- Recharge rate on the top: $q_z = 0.6$ m/yr

- Volumetric flow rate in extraction well: $Q = 20$ gpm

Tasks:

a) Boundary Condition Identification

- Identify and classify each boundary condition shown in the figure
- Explain the difference between Dirichlet and Neumann boundary conditions
- Discuss why each boundary type is appropriate for its respective location

b) Darcy Velocity Calculation

- Calculate the Darcy velocity (q_n) from the aquifer to the stream using the given head measurements
- Show all steps in your calculation including the hydraulic gradient determination

c) Volumetric Discharge Rate

- Calculate the total volumetric discharge rate (Q_n) from the aquifer to the stream
- Express your answer in m^3/day
- Explain the physical meaning of this discharge rate in the context of the overall water balance

Exercise 2: Flow Net Construction and Analysis

Flow nets provide a graphical solution to the Laplace equation for steady-state flow in isotropic and homogeneous media. The figure below shows seepage flow beneath a dam, illustrating the fundamental principles of flow net construction.

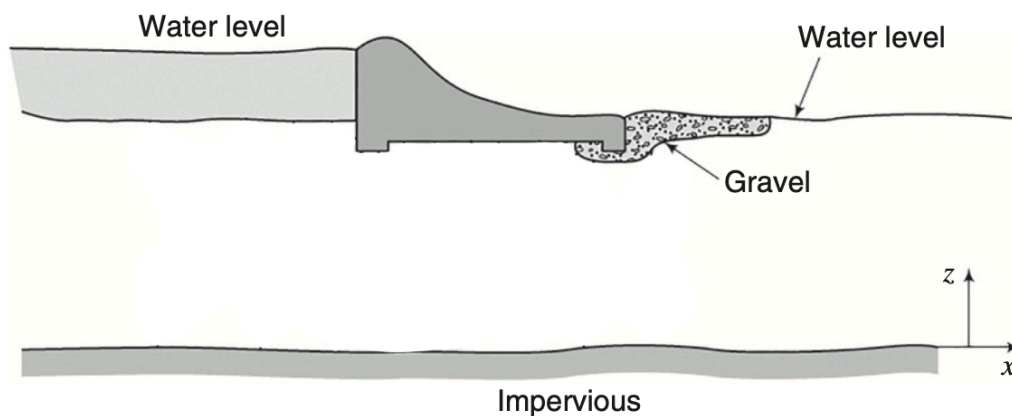


Figure 2: Seepage through a gravel foundation beneath a dam

This flow net demonstrates seepage through a gravel foundation beneath a dam, where flow occurs between the upstream water level (high hydraulic head) and the downstream water level (low hydraulic head). The impervious base provides a no-flow boundary condition.

Key principles for flow net construction:

1. Streamlines are perpendicular to equipotential lines
2. Flow lines and equipotential lines form curvilinear squares when properly constructed
3. The hydraulic head drop between adjacent equipotential lines is constant
4. No flow enters or leaves the system through streamlines

Tasks:

a) **Flow Net Interpretation**

- Identify and count the number of flow channels (stream tubes) in the flow net
- Count the number of equipotential drops between the upstream and downstream boundaries
- Explain why the flow lines curve as they pass under the dam structure

b) **Boundary Condition Analysis**

- Identify the different types of boundary conditions present in this system
- Explain why the bottom boundary (impervious layer) acts as a no-flow boundary
- Describe how the upstream and downstream water levels establish the boundary conditions

c) **Flow Net Properties**

- Explain the concept of “curvilinear squares” and why they are important in flow net construction
- Describe what happens to the flow pattern near the toe of the dam (downstream side)
- Discuss how this flow pattern relates to potential seepage problems or dam stability issues

d) **Practical Applications**

- Explain how this type of flow net analysis would be used in dam design and safety assessment

- Describe what additional information (hydraulic conductivity, geometry) would be needed to calculate actual seepage rates
- Discuss the limitations of 2D flow net analysis for real dam foundations

Exercise 3: Mathematical Analysis of Confined Aquifer Flow

This exercise demonstrates the analytical solution approach for steady-state groundwater flow in a confined aquifer. The figure below shows a one-dimensional flow system between two rivers that penetrate a confined aquifer.

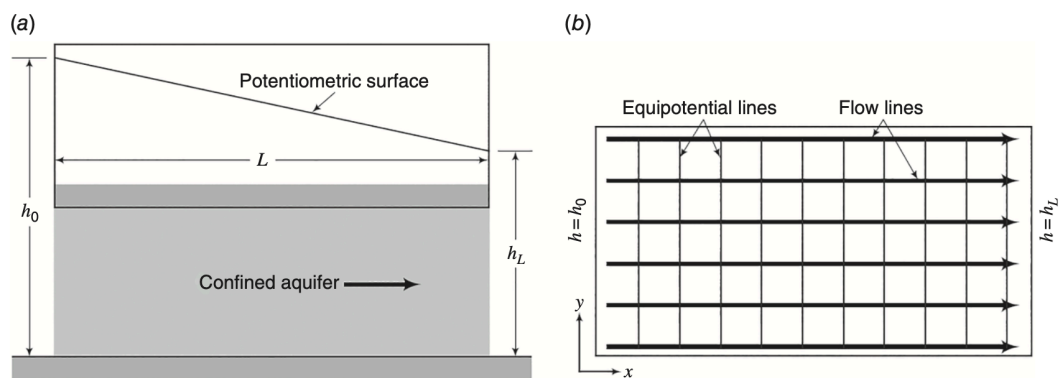


Figure 3: Cross-section and plan view of unidirectional flow in a confined aquifer between two rivers

The system consists of a confined aquifer with constant hydraulic properties, bounded by two rivers that maintain constant head conditions. This creates a one-dimensional steady-state flow problem that can be solved analytically.

Given data from Example 5.5:

- Distance between rivers: $L = 1000$ m
- Aquifer thickness: $b = 20$ m
- Hydraulic conductivity: $K = 20$ m/day
- Hydraulic head at upstream river: $h_0 = 500$ m
- Hydraulic head at downstream river: $h_L = 495$ m
- River width: $W = 600$ m

Governing equation: For one-dimensional steady-state flow in a homogeneous confined aquifer: $\frac{\partial^2 h}{\partial x^2} = 0$

Boundary conditions:

- At $x = 0$: $h = h_0 = 500$ m
- At $x = L$: $h = h_L = 495$ m

Tasks:

a) **Analytical Solution Development**

- Derive the analytical solution for hydraulic head as a function of position
- Show that the solution is: $h = h_0 + (h_L - h_0)\frac{x}{L}$
- Explain why this represents a linear head distribution

b) **Darcy Velocity Calculation**

- Calculate the Darcy flow velocity using $q_x = -K \frac{\partial h}{\partial x}$
- Determine the numerical value of q_x in m/day
- Explain the physical meaning of the negative sign in Darcy's law

c) **Flow Rate Analysis**

- Calculate the total volumetric flow rate Q between the rivers
- Express your answer in m^3/day and m^3/yr
- Compare this flow rate to typical river discharge values

d) **Hydraulic Head Distribution**

- Calculate the hydraulic head at the midpoint between the two rivers ($x = 500$ m)
- Verify this result using the analytical solution
- Sketch the head distribution along the flow path

e) **Sensitivity Analysis**

- Discuss how doubling the hydraulic conductivity would affect the flow rate
- Explain what would happen if pumping wells were installed between the rivers
- Describe the limitations of this one-dimensional analytical approach

Exercise 4: Unconfined Aquifer Flow Analysis (Dupuit Assumption)

This exercise demonstrates groundwater flow analysis in an unconfined aquifer using Dupuit's assumption. Unlike confined aquifers, unconfined aquifers have a free water table surface that changes with the hydraulic head, creating a nonlinear flow problem.

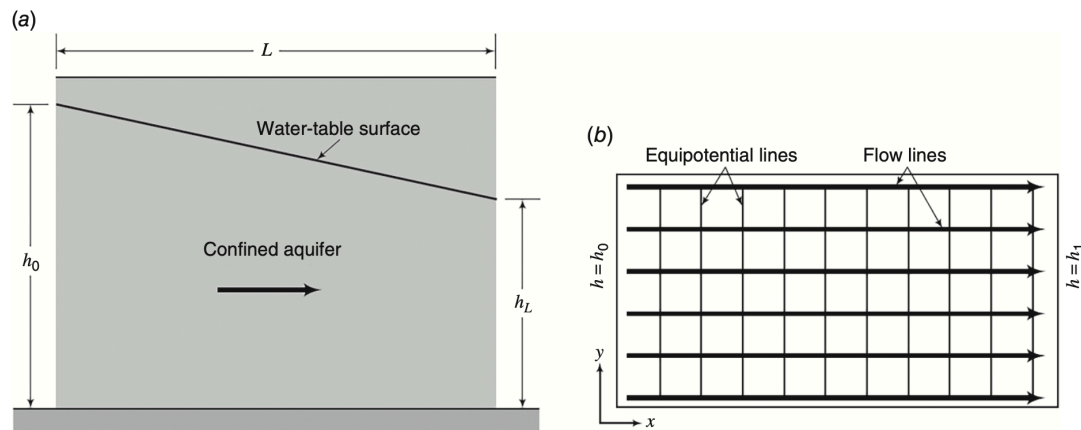


Figure 4: Cross-section and plan view of unidirectional flow in an unconfined aquifer between two rivers with water table surface

The system consists of an unconfined aquifer between two rivers with different water levels. The water table surface slopes from the upstream to downstream river, and flow occurs through the saturated thickness below the water table.

Given data from Example 5.6:

- Distance between rivers: $L = 1000$ m
- Hydraulic conductivity: $K = 20$ m/day
- Hydraulic head at upstream river: $h_0 = 500$ m
- Hydraulic head at downstream river: $h_L = 495$ m
- River width: $W = 600$ m

Governing equation for unconfined flow: For one-dimensional steady-state flow in a homogeneous unconfined aquifer using Dupuit's assumption: $\frac{\partial}{\partial x} \left(h \frac{\partial h}{\partial x} \right) = 0$

Dupuit's assumptions:

1. Flow is horizontal and uniform over the vertical cross-section
2. Hydraulic gradient equals the slope of the water table
3. Vertical flow components are negligible

Boundary conditions:

- At $x = 0$: $h = h_0 = 500$ m
- At $x = L$: $h = h_L = 495$ m

Tasks:**a) Analytical Solution Development**

- Derive the analytical solution starting from $\frac{\partial}{\partial x} \left(h \frac{\partial h}{\partial x} \right) = 0$
- Show that the solution is: $h = \sqrt{h_0^2 + (h_L^2 - h_0^2) \frac{x}{L}}$
- Explain why this creates a curved water table profile (unlike the linear profile in confined aquifers)

b) Hydraulic Head at Midpoint

- Calculate the hydraulic head at $x = 500$ m using the analytical solution
- Compare this result to the confined aquifer case (497.5 m from Exercise 3)
- Explain the physical reason for the difference

c) Darcy Velocity Analysis

- Derive the expression for Darcy velocity: $q = K \frac{h_0^2 - h_L^2}{2Lh}$
- Calculate the Darcy velocity at $x = 500$ m
- Explain why the velocity varies with position in unconfined aquifers

d) Flow Rate Calculation

- Calculate the total flow rate using: $Q_{unitwidth} = K \frac{h_0^2 - h_L^2}{2L}$
- Determine the total volumetric flow rate for the given river width
- Convert your answer to m^3/yr and compare with the confined aquifer result

e) Comparison and Physical Interpretation

- Compare the flow characteristics between confined and unconfined aquifers
- Discuss the validity of Dupuit's assumptions for this problem
- Explain when these assumptions would break down in practice